

Cyrus Guided Notes: MIT 6.006 Introduction to Algorithms

MIT course pack

2026-05-15

Course source

Official source: <https://ocw.mit.edu/courses/6-006-introduction-to-algorithms-fall-2011/>

Why this course is in the system. Algorithm foundation for SLAM graph search, data structures, complexity, and validation scripts.

Learning output. Write a graph, search, and heap formula page, then connect it to simctl data structures.

Prerequisite checkpoint

- Discrete math
- Python or pseudocode
- Proof by induction

Formula checkpoint

Do not memorize these first. Read each formula as a sentence: what changes, what is observed, what is optimized, and what output it produces.

$$\begin{aligned}T(n) &= aT(n/b) + f(n) \\d[v] &= \min_{(u,v) \in E} \{d[u] + w(u,v)\} \\O((V + E) \log V) \\ \text{heapify} &= O(\log n)\end{aligned}$$

Visual page

Draw a graph search frontier and show why a priority queue changes the exploration order.

GoodNotes pages

- Page MIT-A001: asymptotic notation

- Page MIT-A002: graph search and Dijkstra

Web interaction

A draggable graph where edge weights update the shortest-path table.

Obsidian and Notion

Layer	Output
Obsidian	MIT EECS -> 6.006 algorithms
Notion	Solved problem evidence and algorithm notes.

First study move

Open the official source, copy the prerequisite checkpoint into GoodNotes, then write one formula in your own words before watching or reading more.